

EXPLICIT PRIME DENSITIES FOR LUCAS SEQUENCE RANK DIVISIBILITY

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ABSTRACT. Let U be a Lucas sequence, p be a prime, and $\rho_U(p)$ be the rank of appearance of p in U , that is, the least positive integer n such that $p \mid U_n$, if it exists. We derive closed-form formulas for the Dirichlet density of primes p for which $d \mid \rho_U(p)$, where $d \geq 1$ is a fixed integer. Our results complete the work of Sanna (2022) by covering all U and all $d \geq 1$.

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1. Introduction

A *Lucas sequence* $U = U(a_1, a_2) = (U_n)_{n \geq 0}$ with non-zero parameters $a_1, a_2 \in \mathbb{Z}$ is defined by $U_0 = 0$, $U_1 = 1$, and $U_{n+2} = a_1 U_{n+1} - a_2 U_n$ for all $n \geq 0$. Given a prime number p , we denote by $\rho_U(p)$ the smallest positive n at which $p \mid U_n$. This integer is called the *rank of appearance* of p in U and exists for all $p \nmid a_2$. In this paper, we study the Dirichlet density of the set $\mathcal{R}_U(d)$ of primes $p \nmid a_2$ such that $\rho_U(p)$ is divisible by a fixed integer $d \geq 1$. This was first studied by Hasse [7, 8] for sequences $U(a+1, a)$ with $a \in \mathbb{Z}$ square-free, $|a| \geq 2$, and d a prime number. He settled the existence and found an explicit formula for the Dirichlet density. For instance, Hasse found a density of $17/24$ in the case $|a| = d = 2$, and $d^2/(d^2 - 1)$ otherwise. Following his work, many authors considered the problem of determining the density for other Lucas sequences and values of d .

Let $f_U(X) = X^2 - a_1 X + a_2$ be the characteristic polynomial of U . The problem was completely solved by Wiertelak [20, 21, 22] when $f_U(X)$ is reducible. In 1985, Lagarias [11] computed the Dirichlet density of $\mathcal{R}_U(2)$ for three Lucas sequences with irreducible

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polynomial. In particular, he proved that $\mathcal{R}_F(2)$ has density $2/3$, where $F = U(1, -1)$ is the Fibonacci sequence. Later on, the Fibonacci case was completed for all $d \geq 1$ by Cubre and Rouse [4] through the study of algebraic groups. More recently, Sanna [15] made an important contribution to this problem by computing the Dirichlet density for all odd $d \geq 1$ with $3 \nmid d$ if the splitting field of f_U is $\mathbb{Q}(\sqrt{-3})$. This was done using a method of Moree [13]. Prior to Sanna's result, Ballot [1] considered the case of the splitting field being $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, and $d \in \{2, 4\}$ or $d \in \{3, 6\}$ respectively. We want to stress that a cyclotomic splitting field causes many problems when d is divisible by one of 2 or 3.

Let $a, b \in \bar{\mathbb{Q}}$ be the roots of f_U and $K = \mathbb{Q}(a)$. We denote by $\Delta = a_1^2 - 4a_2$ the discriminant of f_U and $\gamma = a/b$ its root quotient. Then, the set $\mathcal{R}_U(d)$ is equal, up to finitely many exceptions, to

$$\mathcal{R}_\gamma(d) = \{p : p \nmid 2a_2\Delta \text{ and } d \mid \text{ord}_\pi(\gamma) \text{ for all } \pi \in \mathcal{O}_K \text{ and } p = \pi \cap \mathbb{Z}\},$$

where $\mathcal{O}_K$ is the ring of integers of K and $\text{ord}_\pi(\gamma)$ is the order of $\gamma \bmod \pi$ in the multiplicative group $(\mathcal{O}_K/\pi)^\times$. Indeed, we have $\rho_U(p) = \text{ord}_\pi(\gamma)$, by [15, Lemma 4.1]. The goal of our paper is to find the Dirichlet density of $\mathcal{R}_U(d)$, if it exists, in all the cases left by Sanna's theorem. Moreover, we want to write it in a closed-form formula, meaning that only finitely many operations are required to compute its value. By the above, the sets $\mathcal{R}_U(d)$ and $\mathcal{R}_\gamma(d)$ have the same Dirichlet density, when it exists. Therefore, we study the set $\mathcal{R}_\gamma(d)$, which is more suitable for some of our calculations.

In the statement of [15, Theorem 1.1], Sanna assumed that d is odd and not divisible by 3 when $\Delta_K = -3$. However, these restrictions are not used in the proof of the existence of the density, nor in the proof of the error term in the asymptotic formula. Indeed, [15, Lemma 5.1] is stated without them and, while they appear in the statement of [15, Lemma 5.3], the proof does not invoke them. The proof of the main theorem only relies on these lemmas and [15, Lemmas 5.2 and 5.4]. It is in the latter that the assumption on d is required to compute a closed-form formula of the density. Let $x > 1$ and denote by $\mathcal{R}_\gamma(d; x)$ the number of primes $p \in \mathcal{R}_\gamma(d)$ with $p \leq x$. Let $K_{n,d} = K(\zeta_n, \gamma^{1/d})$, where $d \mid n$ are integers and ζ_n is a primitive n -th root of unity. We write Li , ω , and φ for the logarithmic integral, the number of distinct prime divisors, and the Euler totient functions respectively. With a slight change of the proof of [15, Lemma 5.2], the result [15, Theorem 1.1] can be restated, with our notation, as follows.

Theorem 1. *Let d be an integer. There exists an absolute constant $B > 0$, such that for every $x > \exp(Bd^{40})$, we have*

$$\mathcal{R}_\gamma(d; x) = \delta_\gamma(d) \text{Li}(x) + \mathcal{O}_\gamma \left(\frac{d}{\varphi(d)} \cdot \frac{x(\log \log x)^{\omega(d)}}{(\log x)^{9/8}} \right),$$

where

$$(1) \quad \delta_\gamma(d) = \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)(1 + [\sigma_{u,v} \text{ exists}])}{[K_{dv,uv} : \mathbb{Q}]},$$

and $[\sigma_{u,v} \text{ exists}]$ is a Boolean that checks the existence of a $\sigma_{u,v}$ in $\text{Gal}(K_{dv,uv}/\mathbb{Q})$ satisfying $\sigma_{u,v}(\sqrt{\Delta}) = -\sqrt{\Delta}$, $\sigma_{u,v}(\zeta_{dv}) = \zeta_{dv}^{-1}$, and $\sigma_{u,v}(\gamma^{1/uv}) = \gamma^{-1/uv}$.

In this paper, we compute closed-form formulas for the series defined in (1) in all the cases left by Sanna [15]. This includes the more difficult cases of a cyclotomic splitting field K discussed above. We note that a particular case of our results was obtained independently by Luo, Hong, and Liu [12]. They extended Sanna's result to even integers $d \geq 2$ under the assumption that K is not cyclotomic and γ is not an n -th power in K for all $n \geq 2$. They used a method similar to Cubre and Rouse [4]. To simplify our calculations, we separate $\delta_\gamma(d)$ into the smaller values

$$(2) \quad \delta_\gamma^+(d) := \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)}{[K_{dv,uv} : \mathbb{Q}]} \quad \text{and} \quad \delta_\gamma^-(d) := \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)[\sigma_{u,v} \text{ exists}]}{[K_{dv,uv} : \mathbb{Q}]},$$

according to the sum $1 + [\sigma_{u,v} \text{ exists}]$. The number $\delta_\gamma^+(d)$ corresponds to the Dirichlet density of $\mathcal{R}_\gamma^+(d)$, the set of primes $p \nmid a_2$ whose rank is divisible by d and with Legendre symbol $(\Delta/p) = 1$. For $\delta_\gamma^-(d)$, we have $(\Delta/p) = -1$ and we call $\mathcal{R}_\gamma^-(d)$ the corresponding set.

In Section 2, we introduce the notation and basic definitions that will be used throughout this paper.

The third section is divided into two parts. First, we give results on powers in cyclotomic extensions of a quadratic field K . For instance, we explicitly construct a 4-th root of a norm 1 element $\gamma \in \mathbb{Q}(i)$ in Lemma 3. Next, we prove several results on *cyclotomic-Kummer extensions* of K . They are extensions of the form $K(\zeta_n, \gamma^{1/d})$, where n and $d \mid n$ are positive integers, and ζ_n is a primitive n -th root of unity. We first compute the degree over $\mathbb{Q}$ of such extensions, and then give necessary and sufficient conditions for special automorphisms to exist in their Galois groups.

Let $\zeta \in K$ be a root of unity and $h(\zeta)$ be the largest $n \geq 1$ such that $\zeta \gamma \in (K^\times)^n$. In Section 4, we compute closed-form formulas for the density defined in (1) in the case where $h(1) \geq h(\zeta)$ for all roots of unity $\zeta \in K$.

In Section 5, we turn to the cases where $h(1)$ is not the maximum over the set of integers $h(\zeta)$ for roots of unity $\zeta \in K$. In these cases, computing $\delta_\gamma(d)$ can be reduced to the computation of $\delta_{\gamma'}(d)$ for some γ' for which $h(1)$ is maximal. In particular, if $h(\zeta)$ is the maximum, then $\gamma' = \zeta\gamma$. Consequently, we find that $\delta_\gamma(d)$ has a closed-form formula for all $d \geq 1$ using Section 4. Thus, these two sections provide a complete solution to the problem of computing $\delta_\gamma(d)$.

Finally, we provide numerical demonstrations of our results in Section 6. For several choices of K , γ and d , we display the value of $\delta_\gamma(d)$ next to its experimental value computed via SageMath [19] programs.

2. Notation

Throughout this paper, we let $\gamma = a/b$ denote the quotient of the roots of the polynomial $f_U(X) := X^2 - a_1X + a_2$. In addition, we assume that γ is not a root of unity, as otherwise one has $U_n = 0$ for infinitely many $n \geq 1$ and the problem becomes trivial. We may also assume that f_U is irreducible since the reducible case was settled by Wiertelak [22]. We let $K = \mathbb{Q}(a)$, $\mu(K)$ be the set of roots of unity contained in K and $\mu_n(K)$ be the set of n -th root of unity in K . Given $\zeta \in \mu(K)$, we let $h(\zeta)$ be the largest integer $n \geq 1$ such that $\zeta\gamma \in (K^\times)^n$. A central object in our paper is

$$h := \max_{\zeta \in \mu(K)} h(\zeta).$$

We use the letters p and d, n for a prime number and positive integers respectively. The p -adic valuation will be denoted by v_p . Given a field L , we denote by Δ_L its absolute discriminant and by $(L^\times)^n$ the subgroup of n -th powers of L . We let d^∞ be the *supernatural number*

$$d^\infty = \prod_{p|d} p^\infty,$$

where we allow prime numbers to have infinite p -adic valuation. To check the validity of a proposition $\mathcal{P}$, we use the Iverson brackets $[\mathcal{P}]$, which is a Boolean function that equals 1 if and only if $\mathcal{P}$ is true. We use ζ_n to denote a primitive n -th root of unity. We denote by (m, n) and $[m, n]$ the gcd and the lcm of integers m and n respectively. The letters φ and μ stand for the Euler totient function and the Möbius function respectively. We denote by $K_{n,d}$ the cyclotomic-Kummer extension $K(\zeta_n, \gamma^{1/d})$ of K when $d \mid n$. Finally, we write $h_m = (h, m^\infty)$ for all $m \geq 1$.

3. Preliminaries on Cyclotomic-Kummer Extensions

This section is divided into two subsections. In the first one, we study whether γ can be expressed as a power in the cyclotomic fields $K(\zeta_n)$. In the second, we compute the degree of cyclotomic-Kummer extensions $K_{n,d} = K(\zeta_n, \gamma^{1/d})$ of K , where $d \mid n$ is a positive integer. In the case $\Delta > 0$, we give necessary and sufficient conditions for the existence of certain automorphisms in their Galois group. When $\Delta < 0$, we show that these automorphisms always exist. Below, we display two theorems found in literature that we use on many occasions.

Theorem 2. *Let K be a field and $a \in K^\times$. Then, $X^n - a$ is irreducible over K if and only if $a \notin (K^\times)^p$ for all primes $p \mid n$ and $a \notin -4(K^\times)^4$ if $4 \mid n$.*

Proof. See [10, Chapter 8, Theorem 1.6]. $\square$

Theorem 3. *Let K be a number field and $m = \#\mu_n(K)$. Then, for all $a \in K$, the extension $K(\zeta_n, a^{1/n})/F$ is abelian if and only if $a^m \in (K^\times)^n$.*

Proof. See [10, Chapter 8, Theorem 3.2]. $\square$

3.1. Powers in cyclotomic extensions

Throughout this subsection, we consider an arbitrary $\gamma \in K$. We determine at which condition a non-square γ becomes a square in $K(\zeta_n)$, where $n \geq 1$. We also study the fourth roots of γ when $\Delta_K = -4$, and its cube roots when $\Delta_K = -3$.

Lemma 4. *Let $\gamma \in K \setminus (\mathbb{Q} \cup K^\times)^2$ with $|N_{L/K}(\gamma)| = 1$ and write $\gamma = u + v\sqrt{\Delta_K}$, where $u, v \in \mathbb{Q}^\times$. Letting $c = (u - 1)/2$, define*

$$K_1 = \mathbb{Q}(\sqrt{c}) \quad \text{and} \quad K_2 = \mathbb{Q}(\sqrt{c/\Delta_K})$$

and let Δ_1 and Δ_2 be their respective absolute discriminants. Then, $\sqrt{\gamma} \in K(\zeta_n)$ if and only if $N_{K/\mathbb{Q}}(\gamma) = 1$, and $\Delta_1 \mid n$ or $\Delta_2 \mid n$.

Proof. First, assume that K is one of $\mathbb{Q}(i)$ or $\mathbb{Q}(\zeta_3)$. In that case, the norm can only be equal to 1. The polynomial $X^4 - 2uX^2 + 1$ annihilates $\sqrt{\gamma}$ and is irreducible over $\mathbb{Q}$ because $\gamma \notin (K^\times)^2$. It has Galois group $C_2 \times C_2$ by [16, Lemma 4.2]. The proof follows as [16, Lemma 4.6], which also deals with the non-cyclotomic cases, and we use [17, Lemma 3] for the condition on the discriminants. $\square$

Lemma 5. *Assume that $\Delta_K = -4$. Let $\gamma \in K \setminus (\mathbb{Q} \cup (K^\times)^2)$ with $N_{K/\mathbb{Q}}(\gamma) = 1$ and write $\gamma = u + v\sqrt{-4}$, where $u, v \in \mathbb{Q}^\times$. Then, a fourth root of γ is given by*

$$(3) \quad \gamma^{1/4} = \left(\frac{1 + \sqrt{c}}{2} \right)^{1/2} + \frac{v\sqrt{-4}}{2|v|} \left(\frac{1 - \sqrt{c}}{2} \right)^{1/2},$$

where $c = (u - 1)/2$. Moreover, the field $L = K(\gamma^{1/4})$ is abelian over $\mathbb{Q}$ and its conductor has the form $\mathfrak{f}(L) = 2^a p_1 \cdots p_s$, where $a \in \{2, 3, 4\}$ and the p_i 's are the odd primes that ramify in L .

Proof. Let z be the right-hand side of (3). We have

$$z^2 = \sqrt{c} + \frac{v\sqrt{-4}}{2|v|} \cdot \sqrt{1 - c} = \sqrt{c} + \frac{v\sqrt{-4}}{2\sqrt{c}},$$

where we used that $c + v^2/c = 1$, because z has modulus 1. We find γ by squaring a second time and using $c - v^2/c = u$. Next, write $L = \mathbb{Q}(i, \gamma^{1/4})$. One can show that a field automorphism of $L/\mathbb{Q}$ is entirely determined by the relations

$$\sigma_{j,k} : \begin{cases} i & \mapsto ij, \\ \gamma^{1/4} & \mapsto i^k \gamma^{j/4}, \end{cases}$$

where $j \in \{-1, 1\}$ and $0 \leq k \leq 3$. We used $N_{K/\mathbb{Q}}(\gamma) = 1$, that is, $\bar{\gamma} = \gamma^{-1}$, to obtain the relations. We can see that they commute, and thus $L/\mathbb{Q}$ is abelian. Finally, we may write $L = \mathbb{Q}(i, \alpha)$, where $2\alpha^2 = 1 + \sqrt{c}$. Therefore, L is the compositum of K and $F = \mathbb{Q}(\alpha)$. By [5, Chapter 2, Proposition 4.1.1], we have

$$\mathfrak{f}(L) = [\mathfrak{f}(K), \mathfrak{f}(F)] = [4, \mathfrak{f}(F)].$$

Next, the minimal polynomial of α is $X^4 - X^2 + (1 - c)/4$. By [16, Lemma 4.2], we see that $F/\mathbb{Q}$ has Galois group C_4 . Hence $\mathfrak{f}(F) = 2^b p_1 \cdots p_s$, where $b \in \{0, 2, 3\}$, by the formula proved by Spearman and Williams [18]. $\square$

Lemma 6. *Assume that $\Delta_K = -3$. Let $\gamma \in K \setminus (\mathbb{Q} \cup (K^\times)^3)$ with $N_{K/\mathbb{Q}}(\gamma) = 1$ and write $\gamma = u + v\sqrt{-3}$, where $u, v \in \mathbb{Q}^\times$. Then, a cube root of γ is given by*

$$(4) \quad \gamma^{1/3} = \frac{r}{2} + \frac{v}{r^2 - 1} \sqrt{-3},$$

where r is a root of $R(X) = X^3 - 3X - 2u$, which is irreducible over $\mathbb{Q}$, has three real roots, and has abelian splitting field L . Moreover, if $p_1 < \cdots < p_s$ denote the ramified primes in L outside of 3, then the conductor of L is of the form

$$\mathfrak{f}(L) = 3^a p_1 \cdots p_s,$$

where $a \in \{0, 2\}$, and $\gamma^{1/3} \in K(\zeta_n)$ if and only if $\mathfrak{f}(L) \mid n$.

Proof. The formula for $\gamma^{1/3}$ was obtained by Cavallo [2] for real quadratic fields. It turns out that the same formula holds in our case. Let z denote the right-hand side of (4). Note that $x + y\sqrt{-3}$, where $x, y \in \mathbb{Q}$, is a cube root of γ if and only if

$$x(x^2 - 9y^2) = u \quad \text{and} \quad 3y(x^2 - y^2) = v.$$

Here, $x = r/2$ and $y = v/(r^2 - 1)$. Let us first compute y^2 . Since γ has norm 1, we have $u^2 + 3v^2 = 1$ and

$$y^2 = \frac{v^2}{(r^2 - 1)^2} = \frac{1 - u^2}{3(r^2 - 1)^2} = \frac{4 - (r^3 - 3r)^2}{12(r^2 - 1)^2},$$

where we used $2u = r^3 - 3r$. Then, expanding the numerator, we obtain

$$y^2 = \frac{-r^6 + 6r^4 - 9r^2 + 4}{12(r^2 - 1)^2} = \frac{-(r^2 - 1)^2(r^2 - 4)}{12(r^2 - 1)^2} = \frac{4 - r^2}{12}.$$

Replacing the value of y^2 in both $x(x^2 - 9y^2)$ and $3y(x^2 - y^2)$ yields u and v respectively. Hence $z^3 = \gamma$. Next, the discriminant $\text{Disc}(R) = 4 \cdot 27 \cdot (1 - u^2) = (18v)^2$ of $R(X)$ is a positive square. Thus, R has three real roots and $\text{Gal}(L/\mathbb{Q}) \cong C_3$. Finally, we have $K(\gamma^{1/3}) = K(r)$ by (4). Therefore, we have $\gamma^{1/3} \in K(\zeta_n)$ if and only if $K(r) \subset K(\zeta_n)$. This is equivalent to $L = \mathbb{Q}(r)$ being a subfield of $\mathbb{Q}(\zeta_n)$ since $\mathbb{Q}(r)$ is totally real. By definition of the conductor, we obtain $\gamma^{1/3} \in K(\zeta_n)$ if and only if $\mathfrak{f}(L) \mid n$. The formula for the conductor of a cubic field was computed by Hasse [6]. (See also the work of Huard, Spearman, and Williams [9] for another characterization.) $\square$

3.2. Results on cyclotomic-Kummer extensions

We assume that $h = h(1)$ and write $\gamma = \gamma_0^h$ for some $\gamma_0 \in K$. We study the cyclotomic-Kummer extensions $K_{n,d} = K(\zeta_n, \gamma^{1/d})$, where $d \mid n$ is a positive integer. In the next theorem, we compute the minimal polynomial of $\gamma^{1/d}$ over $K(\zeta_n)$, which is essential for the determination of the degree of $K_{n,d}/\mathbb{Q}$. We write $d_0 = d/(d, h)$ and $h_0 = h/(d, h)$ throughout this subsection.

Theorem 7. *The minimal polynomial of $\gamma^{1/d}$ over $K(\zeta_n)$ is $X^{d_0/t} - \gamma_0^{h_0/t}$, where*

$$t := \max(m \mid \#\mu(K) : mh_m \mid d \text{ and } \gamma^{1/mh_m} \in K(\zeta_n)).$$

Proof. We use a slightly different definition of t for the proof. We observe that $mh_m \mid d$ if and only if $m \mid d_0$, and $\gamma^{1/mh_m} \in K(\zeta_n)$ if and only if $\gamma_0^{h_0/m} \in K(\zeta_n)$. In each case, the polynomial $X^{d_0/t} - \gamma_0^{h_0/t}$ annihilates $\gamma^{1/d}$ and we use Theorem 2 to prove its irreducibility. Let p be a prime divisor of d_0/t .

Assume that $v_p(t) < v_p(\#\mu(K))$. Since $p \mid d_0$, we have $ph_p \mid d$. Then, by definition of t and its maximality, we know that $\gamma_0^{h_0/t}$ is not a p -th power in $K(\zeta_n)$.

Assume that $v_p(t) = v_p(\#\mu(K))$. By contradiction, assume $\gamma_0^{h_0/t} \in (K(\zeta_n)^\times)^p$, or equivalently, assume $\gamma_0^{h_0} \in K \cap (K(\zeta_n)^\times)^{p^{1+v_p(t)}}$. This is equivalent to

$$\gamma_0^{h_0 p^{v_p(t)}} = x^{p^{1+v_p(t)}},$$

for some $x \in K$, by Theorem 3, because $p^{v_p(t)}$ is the number of p -th roots of unity in K . It follows that $\gamma_0^{h_0} = \zeta_{p^{v_p(t)}}^k x^p$ for some $k \in \mathbb{Z}$. However, this is a contradiction to the maximality of h if $k \equiv 0 \pmod{p^{v_p(t)}}$, and to $h = h(1)$ otherwise.

Finally, assume that $p = 2$ and 4 divides d_0/t . In particular, we have $4 \mid n$. Thus, if we write $\gamma_0^{h_0} = -4x^4$ for some $x \in K(\zeta_n)$, then $\gamma_0^{h_0} = (2ix^2)^2$ is a square. This is a contradiction to the above. $\square$

Corollary 8. *Let t be defined as in Theorem 7. Then,*

$$[K_{n,d} : \mathbb{Q}] = \frac{d\varphi(n)}{(d,h)t} \cdot \begin{cases} 2, & \text{if } \Delta_K \nmid n; \\ 1, & \text{otherwise.} \end{cases}$$

Proof. We know that $K \subset \mathbb{Q}(\zeta_n)$ if and only if $\Delta_K \mid n$ by [17, Lemma 3]. Therefore, we have $[K(\zeta_n) : \mathbb{Q}] = \varphi(n) \cdot 2^{[\Delta_K \nmid n]}$ and the result follows by Theorem 7. $\square$

In future sections, we use Lemmas 4, 5, and 6 to explicitly determine t , defined in Theorem 7. For instance, if $\Delta_K \notin \{-4, -3\}$, then $\mu(K) = \{\pm 1\}$. By Lemma 4, we have $\gamma^{1/2h_2} \in K(\zeta_n)$ if and only if $N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = 1$ and one of Δ_1 or Δ_2 divides n , where the Δ_i 's are defined in the lemma. Thus, $t = 2$ if $2h_2 \mid d$, $N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = 1$ and one of Δ_1 or Δ_2 divides n , and $t = 1$ otherwise.

Theorem 9. *Assume $\Delta_K > 0$. Let $\sigma : K_{n,d} \rightarrow K_{n,d}$ be such that $\sigma|_{\mathbb{Q}} = \text{id}$ and*

$$\sigma : \begin{cases} \sqrt{\Delta} & \mapsto -\sqrt{\Delta}; \\ \zeta_n & \mapsto \zeta_n^{-1}; \\ \gamma^{1/d} & \mapsto \gamma^{-1/d}. \end{cases}$$

If $2h_2 \nmid d$ or $\gamma^{1/h_2} \notin (K(\zeta_n)^\times)^2$, then σ belongs to $\text{Gal}(K_{n,d}/\mathbb{Q})$ if and only if the two following conditions are satisfied:

- (1) $\Delta_K \nmid n$;
- (2) $h_2 \nmid d$, or $h_2 \mid d$ and $N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = 1$.

If $2h_2 \mid d$ and $\gamma^{1/h_2} \in (K(\zeta_n)^\times)^2$, then σ belongs to $\text{Gal}(K_{n,d}/\mathbb{Q})$ if and only if the two following conditions are satisfied:

- (1) $\Delta_K \nmid n$;
- (2) $\sigma_0(\gamma^{1/2h_2}) = \gamma^{-1/2h_2}$,

where $\sigma_0 \in \text{Gal}(K(\zeta_n)/\mathbb{Q})$ satisfies $\sigma_0(\sqrt{\Delta}) = -\sqrt{\Delta}$ and $\sigma_0(\zeta_n) = \zeta_n^{-1}$.

Proof. From the proof of [15, Lemma 4.2], we know σ_0 exists if and only if $\Delta_L \nmid n$. Since $\sigma|_{K(\zeta_n)} = \sigma_0$, it suffices to find necessary and sufficient conditions for σ_0 to be extended into σ . By Theorem 7, let $g(X) = X^{d_0/t} - \gamma_0^{h_0/t}$ be the minimal polynomial of $\gamma^{1/d}$ over $K(\zeta_n)$. Since $K_{n,d} \cong K(\zeta_n)[X]/(g(X))$, we can extend σ_0 by sending a root of g to any root of $\sigma_0 g$. Equivalently, we need

$$(5) \quad \sigma_0(\gamma_0^{h_0/t}) = \gamma_0^{-h_0/t}.$$

If $t = 2$, then we show that this is equivalent to $\sigma_0(\gamma^{1/2h_2}) = \gamma^{-1/2h_2}$. The direct implication is trivial by raising (5) to the (d, h') -th power, where $h' = h/h_2$. For the

converse, taking the (d, h') -th root on both sides, we obtain

$$\sigma_0(\gamma^{1/2h_2(d, h')}) = \sigma_0(\gamma_0^{h_0/2}) = \zeta_{(d, h')}^k \gamma_0^{-h_0/2},$$

for some $k \in \mathbb{Z}$. Squaring both sides, we have $\sigma_K(\gamma_0^{h_0}) = \zeta_{(d, h')}^{2k} \gamma_0^{-h_0}$, which holds in K . Hence $\zeta_{(d, h')}^{2k} = 1$ and, because $2 \nmid (d, h')$, we have $\zeta_{(d, h')}^k = 1$ as well.

If $t = 1$, then we have two cases. If $h_2 \mid d$, then $\gamma_0^{h_0}$ is a square in K . Moreover, because $\sigma_K(\gamma) = \gamma^{-1}$, we have $\sigma_K(\gamma_0^{h_0/2}) = \pm \gamma_0^{-h_0/2}$ and it follows that $\sigma_0(\gamma_0^{h_0}) = \gamma_0^{-h_0}$ by squaring both sides. If $h_2 \nmid d$, then $\sigma_0(\gamma_0^{h_0}) = \gamma_0^{-h_0}$ if and only if $\sigma_K(\gamma^{1/h_2}) = \gamma^{-1/h_2}$. This is equivalent to the norm of γ^{1/h_2} being 1. $\square$

We end this section with a short lemma on the existence of the automorphism $\sigma_{u,v}$ defined in Theorem 1. This will ease calculations when $\Delta < 0$, particularly when K is a cyclotomic field. Indeed, a direct consequence is that $\delta_\gamma(d) = 2\delta_\gamma^+(d)$.

Lemma 10. *Assume that $\Delta_K < 0$. Then, the Galois group of $K_{dv, uv}/\mathbb{Q}$ contains an automorphism $\sigma_{u,v}$ as defined in Theorem 1 for all $u \mid d$ and $v \mid d^\infty$.*

Proof. Let τ be the complex conjugation. Because $K_{dv, uv}/\mathbb{Q}$ is a finite Galois extension, the restriction $\tau|_{K_{dv, uv}}$ is a field automorphism of $K_{dv, uv}$. Since $\gamma \in K$ and $\Delta_K < 0$, we have $\tau(\gamma) = \gamma^{-1}$. Hence $\tau(\gamma^{1/uv}) = \gamma^{-1/uv}$ and $\sigma_{u,v} = \tau|_{K_{dv, uv}}$. $\square$

4. The case $h = h(1)$

In this section, we derive closed-form formulas for $\delta_\gamma(d)$ in the case $h = h(1)$. To do so, we compute $\delta_\gamma^+(d)$ and $\delta_\gamma^-(d)$ separately. Recall that $\gamma = \gamma_0^h$ for some $\gamma_0 \in K$ and that $h_m = (h, m^\infty)$ for all $m \geq 2$. We define the Boolean variable

$$\mathcal{Q} = [N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = 1].$$

We first study the case $\Delta_K \notin \{-4, -3\}$, which has two subcases: when $\mathcal{Q} = 0$, which forces $\Delta_K > 0$, and when $\mathcal{Q} = 1$. Then, we consider the case $\Delta_K = -4$, and the case $\Delta_K = -3$. We start with the following lemma that generalizes a formula given in the proof of [15, Lemma 5.4]:

Lemma 11. *Let $d, h \geq 1$, $e \mid d^\infty$ and $\nu \mid d^\infty$ be positive integers and define*

$$(6) \quad S_{d,e,h}(\nu) := \sum_{\substack{v \mid d^\infty \\ e \mid v}} \sum_{u \mid d} \frac{\mu(u)(uv, h)[\nu \mid uv]}{\varphi(dv)uv}.$$

If $d := D(d, \nu^\infty)$ and $(h, \nu^\infty) \mid \nu$, then

$$S_{d,e,h}(\nu) = \frac{(h, d^\infty)\nu}{d\varphi(\nu)[e, \nu(h, D^\infty)]^2} \cdot \prod_{p \mid \nu} \left(1 - \frac{(pe, \nu)^2}{p(e, \nu)^2}\right) \cdot \prod_{p \mid d} \left(\frac{p^2}{p^2 - 1}\right).$$

Proof. For $\nu = 1$, see the proof of [15, Lemma 5.4]. Let us write

$$u = u_1 u_2, \quad v = v_1 v_2 \quad \text{and} \quad e = e_1 e_2,$$

where $u_2 = (u, \nu^\infty)$, $v_2 = (v, \nu^\infty)$, and $e_2 = (e, \nu^\infty)$. We have $[\nu \mid uv] = [\nu \mid u_2 v_2]$. Using $\varphi(dv) = \varphi(d)v$ and the multiplicativity of the Möbius function μ , the Euler totient φ and the gcd, we obtain

$$S_{d,e,h}(\nu) = \sum_{\substack{v_1 \mid D^\infty \\ e_1 \mid v_1}} \sum_{\substack{v_2 \mid \nu^\infty \\ e_2 \mid v_2}} \sum_{\substack{u_1 \mid D \\ u_2 \mid \nu}} \left(\frac{\mu(u_1)(u_1 v_1, h)}{\varphi(D)u_1 v_1^2} \cdot \frac{\mu(u_2)(u_2 v_2, h)[\nu \mid u_2 v_2]}{\varphi((d, \nu^\infty))u_2 v_2^2} \right).$$

Hence $S_{d,e,h}(\nu) = S_{D,e_1,h}(1) \cdot S_{(d, \nu^\infty), e_2, h}(\nu)$. It remains to compute $S_{(d, \nu^\infty), e_2, h}(\nu)$. Since $(h, \nu^\infty) \mid \nu$, we have

$$S_{(d, \nu^\infty), e_2, h}(\nu) = \frac{(h, \nu^\infty)}{\varphi((d, \nu^\infty))} \sum_{\substack{v_2 \mid \nu^\infty \\ e_2 \mid v_2}} \sum_{u_2 \mid \nu} \frac{\mu(u_2)[\nu \mid u_2 v_2]}{u_2 v_2^2},$$

as $\nu \mid u_2 v_2$ implies that $(u_2 v_2, h) = (h, \nu^\infty)$. We may now interchange the sum and the series to obtain

$$S_{(d, \nu^\infty), e_2, h}(\nu) = \frac{(h, \nu^\infty)}{\varphi((d, \nu^\infty))} \sum_{u_2 \mid \nu} \frac{\mu(u_2)}{u_2} \sum_{\substack{v_2 \mid \nu^\infty \\ \ell \mid v_2}} \frac{1}{v_2^2},$$

where $\ell = [e_2, \nu/u_2]$. The inner sum is computed using the Euler product formula:

$$\sum_{\substack{v_2 \mid \nu^\infty \\ \ell \mid v_2}} \frac{1}{v_2^2} = \frac{1}{\ell^2} \sum_{v_2 \mid \nu^\infty} \frac{1}{v_2^2} = \frac{1}{\ell^2} \prod_{p \mid \nu} \sum_{r \geq 0} p^{-2r} = \frac{1}{\ell^2} \prod_{p \mid \nu} \left(\frac{p^2}{p^2 - 1} \right).$$

We obtain

$$S_{(d, \nu^\infty), e_2, h}(\nu) = \frac{(h, \nu^\infty)}{\varphi((d, \nu^\infty))} \prod_{p \mid \nu} \left(\frac{p^2}{p^2 - 1} \right) \cdot \sum_{u_2 \mid \nu} \frac{\mu(u_2)}{u_2 [e_2, \nu/u_2]^2}.$$

To compute the remaining sum, call it S , we expand $[e_2, \nu/u_2]$ and use properties of the gcd. We find that

$$S = \sum_{u_2 \mid \nu} \frac{\mu(u_2)(u_2 e_2, \nu)^2}{u_2 (e_2 \nu)^2} = \frac{(e_2, \nu)^2}{(e_2 \nu)^2} \sum_{u_2 \mid \nu} \frac{\mu(u_2)(u_2 e_2, \nu)^2}{u_2 (e_2, \nu)^2}.$$

The general term of the sum is a multiplicative function in u_2 , so we apply the Euler product formula again and find

$$S = \frac{(e_2, \nu)^2}{(e_2 \nu)^2} \prod_{p|\nu} \left(1 - \frac{(pe_2, \nu)^2}{p(e_2, \nu)^2} \right) = \frac{(e, \nu)^2}{(e_2 \nu)^2} \prod_{p|\nu} \left(1 - \frac{(pe, \nu)^2}{p(e, \nu)^2} \right),$$

where we used that $e_2 = (e, \nu^\infty)$ on the second equality. Next, applying [15, Lemma 5.4] to $S_{D, e_1, h}(1)$, we find that

$$S_{d, e, h}(\nu) = A \cdot \prod_{p|d} \left(\frac{p^2}{p^2 - 1} \right) \cdot \prod_{p|\nu} \left(1 - \frac{(pe, \nu)^2}{p(e, \nu)^2} \right),$$

where A is defined by

$$A := \frac{(h, D^\infty)}{D[(h, D^\infty), e_1]^2} \cdot \frac{(h, \nu^\infty)(e, \nu)^2}{\varphi((d, \nu^\infty))(e_2 \nu)^2} = \frac{(h, d^\infty)\nu}{d\varphi(\nu)} \cdot \frac{(e, \nu)^2}{(e_2 \nu)^2 [(h, D^\infty), e_1]^2}.$$

We used the following identities in the second equality:

$$\frac{(d, \nu^\infty)}{\varphi((d, \nu^\infty))} = \frac{\nu}{\varphi(\nu)} \quad \text{and} \quad (h, D^\infty)(h, \nu^\infty) = (h, d^\infty).$$

Finally, we can expand the lcm to find

$$\frac{(e, \nu)^2}{(e_2 \nu)^2 [(h, D^\infty), e_1]^2} = \frac{(e, \nu)^2 (h, e_1, D^\infty)^2}{(e_2 \nu)^2 (h, D^\infty)^2 e_1^2} = \frac{(e, \nu)^2 (h, e, D^\infty)^2}{\nu^2 (h, D^\infty)^2 e^2}.$$

Since ν and (h, D^∞) are coprime, we have $(e, \nu)(h, e, D^\infty) = (e, \nu(h, D^\infty))$. We see that the formula becomes

$$\frac{(e, \nu(h, D^\infty))^2}{\nu^2 (h, D^\infty)^2 e^2} = \frac{1}{[e, \nu(h, D^\infty)]^2},$$

and the result follows. $\square$

Remark 12. Note that $S_{d, eh}(\nu)$, as defined in (6), is zero if $e \nmid d^\infty$ or $\nu \nmid d^\infty$.

In the rest of the paper, our goal is to write $\delta_\gamma^+(d)$ and $\delta_\gamma^-(d)$ as linear combinations of the sums $S_{d, e, h}(\nu)$. Then, the density values will be in closed-forms by Lemma 11 and Remark 12. We write $S_{d, e, h}$ as a shorthand for $S_{d, e, h}(1)$.

4.1. The case $\Delta_K \notin \{-4, -3\}$ and $\mathcal{Q} = 0$

First, note that $h_2 \geq 2$ because $N_{L/\mathbb{Q}}(\gamma) = -1$. Moreover, if $\gamma^{1/h_2} = u + v\sqrt{\Delta_K}$ for some $u, v \in \mathbb{Q}^\times$, then $\mathcal{Q} = 0$ is equivalent to $u^2 - v^2 \Delta_K = -1$. Therefore, the assumption $\mathcal{Q} = 0$ ensures that $2 \mid h_2$ and $\Delta_K > 0$. These facts are used to find a closed-form of $\delta_\gamma^-(d)$.

Theorem 13. Assume $\mathcal{Q} = 0$ and $2 \mid d$. Let $e = \Delta_K / (d, \Delta_K)$. Then,

$$\delta_\gamma^+(d) = \frac{1}{2d} \left(\frac{1}{(h, d^\infty)} + [e \mid d^\infty] \cdot \frac{(h, d^\infty)}{[(h, d^\infty), e]^2} \right) \prod_{p|d} \left(\frac{p^2}{p^2 - 1} \right).$$

Proof. Since $\mathcal{Q} = 0$, and thus $N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = -1$, we see that $\gamma^{1/h_2} \notin (K(\zeta_n)^\times)^2$ by Lemma 4. By Corollary 8, we have

$$(7) \quad [K_{dv,uv} : \mathbb{Q}] = \frac{\varphi(dv)uv}{(uv, h)} \cdot 2^{[\Delta_K \nmid dv]}.$$

Hence, because $\Delta_K \mid dv$ if and only if $e \mid v$, we obtain

$$\delta_\gamma^+(d) = \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)(uv, h)}{\varphi(dv)uv} \cdot \frac{1}{2^{[e \nmid v]}} = \frac{S_{d,1,h} + S_{d,e,h}}{2}.$$

Note that we used the identity $2^{1-[e \nmid v]} = 1 + [e \mid v]$ in the last equality. The result follows by Lemma 11 and Remark 12. $\square$

Theorem 14. Assume $\mathcal{Q} = 0$ and $2 \mid d$. Let $e = \Delta_K / (d, \Delta_K)$. Then,

$$\delta_\gamma^-(d) = \frac{3}{2d} \left(\frac{1}{(h, d^\infty)} - [e \mid d^\infty \text{ and } h_2 \nmid e] \cdot \frac{(h, d^\infty)}{[(h, d^\infty), e]^2} \right) \prod_{p|d} \left(\frac{p^2}{p^2 - 1} \right).$$

Proof. Recall that $\mathcal{Q} = 0$ implies $2 \mid h$ and $\Delta_K > 0$. By Theorem 9, we know that $\sigma_{u,v}$ exists if and only if $h_2 \nmid uv$ and $\Delta_K \nmid dv$. Using Corollary 8, we obtain

$$\delta_\gamma^-(d) = \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)(uv, h)}{\varphi(dv)uv} \cdot \frac{[e \nmid v][h_2 \nmid uv]}{2^{[e \nmid v]}}.$$

We linearize $\delta_\gamma^-(d)$ using

$$\frac{[e \nmid v][h_2 \nmid uv]}{2^{[e \nmid v]}} = \frac{1 - [h_2 \mid uv] - [e \mid v] + [e \mid v][h_2 \mid uv]}{2},$$

so that, with the notation of Lemma 11, we obtain

$$\delta_\gamma^-(d) = \frac{1}{2} \left(S_{d,1,h} - S_{d,1,h}(h_2) - S_{d,e,h} + S_{d,e,h}(h_2) \right).$$

By Lemma 11 with $\nu = h_2$, we see that $S_{d,1,h}(\nu) = -2S_{d,1,h}$. Similarly, when $e \mid d^\infty$, we have $S_{d,e,h}(\nu) = (1 - 3 \cdot [h_2 \nmid e]) \cdot S_{d,e,h}$. Hence

$$\delta_\gamma^-(d) = \frac{3}{2} \left(S_{d,1,h} - [h_2 \nmid e] \cdot S_{d,e,h} \right),$$

and the result follows by Lemma 11 and Remark 12. $\square$

4.2. The case $\Delta_K \notin \{-4, -3\}$ and $\mathcal{Q} = 1$

To compute t (defined in Theorem 7), we have to check whether γ^{1/h_2} is a square or not in $K(\zeta_n)$. Let $\gamma^{1/h_2} = u + v\sqrt{\Delta_K}$, where $u, v \in \mathbb{Q}^\times$. Let $c = (u - 1)/2$, and Δ_1 and Δ_2 be the absolute discriminants of $K_1 = \mathbb{Q}(\sqrt{c})$ and $K_2 = \mathbb{Q}(\sqrt{c/\Delta_K})$ respectively. By Lemma 4, we know that $\gamma^{1/h_2} \in (K(\zeta_n)^\times)^2$ if and only if $\Delta_1 \mid n$ or $\Delta_2 \mid n$. Moreover, we

have

$$(8) \quad \gamma^{1/2h_2} = \sqrt{c} + \frac{v}{2} \sqrt{\frac{\Delta_K}{c}}.$$

Note that $c = 0$ only if $\gamma = 1$, which contradicts that γ is not a root of unity. In addition, we have $[\Delta_K, \Delta_1] = [\Delta_K, \Delta_2] = [\Delta_1, \Delta_2]$ because K , K_1 and K_2 are pairwise linearly disjoint over $\mathbb{Q}$ with $KK_1 = KK_2 = K_1K_2$.

Theorem 15. *Assume $\mathcal{Q} = 1$ and $2 \mid d$. Define*

$$e = \frac{|\Delta_K|}{(d, |\Delta_K|)} \quad \text{and} \quad e_i = \frac{|\Delta_i|}{(d, |\Delta_i|)},$$

for all $1 \leq i \leq 2$. Then,

$$\delta_\gamma^+(d) = \frac{1}{2} \left(S_{d,1,h} + S_{d,e,h} + S_{d,e_1,h}(2h_2) + S_{d,e_2,h}(2h_2) \right).$$

In particular, if $\Delta_K < 0$, then $\delta_\gamma(d) = 2\delta_\gamma^+(d)$.

Proof. By Corollary 8, we have

$$\delta_\gamma^+(d) = \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)(uv, h)}{\varphi(dv)uv} \cdot \frac{t}{2^{[\Delta_K \nmid dv]}}.$$

Then, we linearize $t \cdot 2^{-[\Delta_K \nmid dv]}$ using the identities $2^{1-[\Delta_K \nmid dv]} = 1 + [\Delta_K \mid dv]$ and $t = 1 + [2h_2 \mid uv] \cdot [\Delta_1 \mid dv \text{ or } \Delta_2 \mid dv]$, which is obtained from Lemma 4. Next, we use the inclusion-exclusion principle on t , the equivalence $\Delta_K \mid dv$ if and only if $e \mid v$, which is also valid for (Δ_1, e_1) and (Δ_2, e_2) , and the equalities $[e, e_1] = [e, e_2] = [e_2, e_3]$ discussed at the beginning of Subsection 4.2. We obtain

$$\frac{t}{2^{[\Delta_K \nmid dv]}} = \frac{1 + [e \mid v]}{2} + \frac{[2h_2 \mid uv] \cdot [e_1 \mid v]}{2} + \frac{[2h_2 \mid uv] \cdot [e_2 \mid v]}{2}.$$

We may now linearize $\delta_\gamma^+(d)$ using the above to find the result. Finally, if $\Delta_K < 0$, then $\delta_\gamma^+(d) = \delta_\gamma^-(d)$ by Lemma 10. $\square$

We now turn our attention to $\delta_\gamma^-(d)$ in the case $\Delta_K > 0$. Let us first revise the existence conditions for the automorphism $\sigma \in \text{Gal}(K_{n,d}/\mathbb{Q})$ defined in Theorem 9. Since $\mathcal{Q} = 1$, we see that σ exists if and only if

- (1) $\Delta_K \nmid n$;
- (2) $\mathcal{P}_1(n, d) = 1$ or $\mathcal{P}_2(n, d) = 1$,

with Boolean functions

$$(9) \quad \mathcal{P}_1(n, d) = [2h_2 \nmid d \text{ or } \forall i \in \{1, 2\}, \Delta_i \nmid n],$$

and

$$(10) \quad \mathcal{P}_2(n, d) = [2h_2 \mid d] \cdot [\Delta_1 \mid n \text{ or } \Delta_2 \mid n] \cdot [\sigma_0(\gamma^{1/2h_2}) = \gamma^{-1/2h_2}],$$

where $\sigma_0 \in \text{Gal}(K(\zeta_n)/K)$ is such that $\sigma_0(\sqrt{\Delta}) = -\sqrt{\Delta}$ and $\sigma_0(\zeta_n) = \zeta_n^{-1}$. The next lemma provides a way to compute $\sigma_0(\gamma^{1/2h_2})$ without having to consider $K(\zeta_n)$.

Lemma 16. *Assume $\mathcal{Q} = 1$, $\Delta_K > 0$ and $\Delta_K \nmid n$. Then, we have*

$$\mathcal{P}_2(n, d) = [2h_2 \mid d] \cdot \begin{cases} [\Delta_1 \mid n], & \text{if } c < 0; \\ [\Delta_2 \mid n], & \text{if } c > 0, \end{cases}$$

where c is defined in (8).

Proof. By the proof of [15, Lemma 4.2], we know $\sigma_0 \in \text{Gal}(K(\zeta_n)/K)$ because $\Delta_K \nmid n$. Moreover, we only work on the Boolean variable $[\sigma_0(\gamma^{1/2h_2}) = \gamma^{-1/2h_2}]$ that appears in $\mathcal{P}_2(n, d)$. Thus, we may assume that $2h_2 \mid d$ and one of Δ_1 or Δ_2 divides n . The latter ensures the existence of a square root of γ^{1/h_2} in $K(\zeta_n)$. Note that $[\Delta_1, \Delta_2] \nmid n$, otherwise Δ_K would divide n . From (8), we find

$$\gamma^{-1/2h_1} = -\sqrt{c} + \frac{v}{2} \sqrt{\frac{\Delta_K}{c}}.$$

By comparing them, we see that it suffices to find conditions to have $\sigma_0(\sqrt{c}) = -\sqrt{c}$ and $\sigma_0(\sqrt{c/\Delta_K}) = \sqrt{c/\Delta_K}$.

Assume that $\Delta_1 \mid n$. By definition, we have $\sqrt{c} \in \mathbb{Q}(\zeta_n)$. Since $\sigma_0|_{\mathbb{Q}(\zeta_n)}$ is the complex conjugation, for $\sqrt{c}$ to be sent to $-\sqrt{c}$, we need $c < 0$. We obtain

$$\sigma_0 \left(\sqrt{\frac{c}{\Delta_K}} \right) = \sqrt{\frac{c}{\Delta_K}},$$

since $\sigma_0(\sqrt{\Delta_K}) = -\sqrt{\Delta_K}$. Hence $\sigma_0(\gamma^{1/2h_2}) = \gamma^{-1/2h_2}$. The case $\Delta_2 \mid n$ proceeds similarly, with c/Δ_K instead of c . $\square$

Theorem 17. *Assume $\mathcal{Q} = 1$, $\Delta_K > 0$, and $2 \mid d$. Define*

$$e = \frac{\Delta_K}{(d, \Delta_K)} \quad \text{and} \quad e_i = \frac{|\Delta_i|}{(d, |\Delta_i|)},$$

for all $1 \leq i \leq 2$. Then,

$$\delta_\gamma^-(d) = \frac{S_{d,1,h} - S_{d,e,h}}{2} + (-1)^{[c>0]} \cdot \frac{S_{d,e_1,h}(2h_2) - S_{d,e_2,h}(2h_2)}{2},$$

where c is defined in (8).

Proof. Let $\nu = 2h_2$. By (9) and (10), we have $(\mathcal{P}_1(dv, uv) + \mathcal{P}_2(dv, uv)) \cdot [\Delta_K \nmid dv]$ equals 1 if and only if $\sigma_{u,v}$ exists. We may write $\delta_\gamma^-(d) = S_1(d) + S_2(d)$, where

$$S_i(d) = \sum_{v|d^\infty} \sum_{u|d} \frac{\mu(u)\mathcal{P}_i(dv, uv)}{[K_{dv,uv} : \mathbb{Q}]} \cdot [\Delta_K \nmid dv],$$

for all $i \in \{1, 2\}$. We start with $S_1(d)$. Assuming $\mathcal{P}_1(dv, uv) = 1$ and $\Delta_K \nmid dv$, and using Corollary 8, we obtain a general term for $S_1(d)$ of

$$\frac{\mu(u)(uv, h)}{2\varphi(dv)uv} \cdot \mathcal{P}_1(dv, uv) \cdot [e \nmid v].$$

We used $\Delta_K \mid dv$ if and only if $e \mid v$, which is also valid for (e_1, Δ_1) and (e_2, Δ_2) . Next, we may write $\mathcal{P}_1(dv, uv) = 1 - [2h_1 \mid uv]$ and $\exists i \in \{1, 2\}$, $\Delta_i \mid dv$ and, because of the equalities $[e, e_1] = [e, e_2] = [e_1, e_2]$, we find that

$$\mathcal{P}_1(dv, uv) \cdot [e \nmid v] = (1 - [2h_2 \mid uv] \cdot [e_1 \mid v] - [2h_2 \mid uv] \cdot [e_2 \mid v]) \cdot [e \nmid v].$$

We can now use $[e \nmid v] = 1 - [e \mid v]$ to obtain

$$S_1(d) = \frac{1}{2} \left(S_{d,1,h} - S_{d,e,h} - S_{d,e_1,h}(2h_2) - S_{d,e_1,h}(2h_2) \right) + S_{d,[e_1,e_2],h}(2h_2).$$

We now turn our attention to $S_2(d)$. By Lemma 16, we have

$$\mathcal{P}_2(n, d) \cdot [\Delta_K \nmid dv] = [2h_2 \mid uv] \cdot [\Delta_K \nmid dv] \cdot \begin{cases} [\Delta_1 \mid dv], & \text{if } c < 0; \\ [\Delta_2 \mid dv], & \text{if } c > 0, \end{cases}$$

If $c < 0$, then

$$S_2(d) = \sum_{\substack{v|d^\infty \\ e_1|v}} \sum_{u|d} \frac{\mu(u)(uv, h)t}{2\varphi(dv)uv} \cdot [2h_2 \mid uv] \cdot [e \nmid v] = S_{d,e_1,h}(\nu) - S_{d,[e,e_1],h}(\nu),$$

using Corollary 8 for the field degrees and Lemma 4 to show that $t = 2$, where t is defined in the corollary. Going back to $\delta_\gamma^-(d) = S_1(d) + S_2(d)$, we obtain

$$\delta_\gamma^-(d) = \frac{1}{2} \left(S_{d,1,h} - S_{d,e,h} - S_{d,e_1,h}(\nu) + S_{d,e_2,h}(\nu) \right),$$

where we used $[e, e_1] = [e_1, e_2]$. If $c > 0$, then $S_2(d) = S_{d,e_2,h}(\nu) - S_{d,[e,e_2],h}(\nu)$, and the result follows similarly. $\square$

4.3. The case $\Delta_K = -4$

We now deal with the case of $K = \mathbb{Q}(i)$. Similar to Subsection 4.2, we have to check if γ^{1/h_2} is a square in cyclotomic extensions $K(\zeta_n)$ or not. Because $N_{K/\mathbb{Q}}(\gamma^{1/h_2}) = 1$, it is a square if and only if $\Delta_1 \mid n$ or $\Delta_2 \mid n$, where the Δ_i 's were defined in the previous subsection. However, when it is a square, we also have to check if γ^{1/h_2} is a 4-th power

in $K(\zeta_n)$. By Lemma 5, if we consider $\mathfrak{f}(L)$, the conductor of $L := \mathbb{Q}(i, \gamma^{1/4h_2})$, then $\gamma^{1/4h_2} \in K(\zeta_n)$ if and only if $\mathfrak{f}(L) \mid [4, n]$.

Theorem 18. *Assume that $\Delta_K = -4$ and $2 \mid d$. Define $e = 4/(d, 4)$,*

$$f = \frac{\mathfrak{f}(L)}{(d, \mathfrak{f}(L))} \quad \text{and} \quad e_i = \frac{|\Delta_i|}{(d, |\Delta_i|)},$$

for all $1 \leq i \leq 2$. Then,

$$\delta_\gamma(d) = S_{d,1,h} + S_{d,e,h} + S_{d,e_1,h}(2h_2) + S_{d,e_2,h}(2h_2) + 4S_{d,f,h}(4h_2).$$

Proof. The proof is similar to those of Theorems 13, 14, 15, and 17, thus we only give the main steps. First, we linearize t defined in Theorem 7 as

$$t = 1 + [2h_2 \mid uv] \cdot [\exists i, \Delta_i \mid dv] + 2[4h_2 \mid uv] \cdot [\mathfrak{f}(L) \mid dv]$$

for the field $K_{dv,uv}$. Note that we used $4h_2 \mid uv$ and u being square-free, so that $2 \mid v$. Hence $4 \mid dv$ and $[4, dv] = dv$. Secondly, by Corollary 8, we have

$$\frac{1}{[K_{dv,uv} : \mathbb{Q}]} = \frac{(uv, h)t}{\varphi(dv)uv} \cdot \frac{1}{2^{[4dv]}} = \frac{(uv, h)t}{\varphi(dv)uv} \cdot \frac{1 + [4 \mid dv]}{2}.$$

Finally, replacing this expression in $\delta_\gamma(d)$, we can linearize t and write $\delta_\gamma(d)$ in terms of sums $S_{d,e,h}(\nu)$. Also, note that we use $\Delta_i \mid dv$ if and only if $e_i \mid v$, and the same holds for the pairs $(4, e)$ and $(\mathfrak{f}(L), f)$, and $\delta_\gamma^+(d) = \delta_\gamma^-(d)$ by Lemma 10. $\square$

4.4. The case $\Delta_K = -3$

This is the last case for $h = h(1)$. As before, we check whether γ^{1/h_2} has a square root in $K(\zeta_n)$ or not, using the discriminants Δ_1 and Δ_2 , which we defined as in Subsection 4.2. In addition, we have to check whether γ^{1/h_3} is a cube in $K(\zeta_n)$ or not. Let us write $\gamma^{1/h_3} = u + v\sqrt{-3}$ for some $u, v \in \mathbb{Q}^\times$, and let L be the splitting field of $X^3 - 3X - 2u$. If $\mathfrak{f}(L)$ is the conductor of L , then $\gamma^{1/h_3} \in (K(\zeta_n)^\times)^3$ if and only if $\mathfrak{f}(L) \mid n$, by Lemma 6.

Theorem 19. *Assume that $\Delta_K = -3$ and $(d, 6) > 1$. Define*

$$f = \frac{\mathfrak{f}(L)}{(d, \mathfrak{f}(L))} \quad \text{and} \quad \tilde{e} = \min \left(\frac{|\Delta_i|}{(d, |\Delta_i|)} : 1 \leq i \leq 2 \right).$$

Then, we have

$$\delta_\gamma(d) = 2^{[3|d]} \left(S_{d,1,h} + S_{d,\tilde{e},h}(2h_2) + 2S_{d,f,h}(3h_3) + 2S_{d,[\tilde{e},f],h}(6h_6) \right).$$

Proof. As in the proof of Theorem 18, we may skip a few details. Let t be defined as in Theorem 7 applied to the field $K_{dv,uv}$. Then, we have

$$t = (1 + [2h_2 \mid uv] \cdot [\exists i, \Delta_i \mid dv]) \cdot (1 + 2 \cdot [3h_3 \mid uv] \cdot [\mathfrak{f}(L) \mid dv]),$$

Moreover, the fields $K_1 = \mathbb{Q}(\sqrt{c})$ and $K_2 = \mathbb{Q}(\sqrt{-3c})$ have discriminants Δ_1 and Δ_2 , which are equal up to a factor of 3. If $3 \nmid d$, then it means only one of Δ_1 or Δ_2 may divide dv . If $3 \mid d$, then $\Delta_1 \mid dv$ if and only if $\Delta_2 \mid dv$. In both cases, we obtain

$$[\exists i, \Delta_i \mid dv] = [\tilde{e} \mid v],$$

which simplifies the expression of t . Next, by Corollary 8, we have

$$[K_{dv,uv} : \mathbb{Q}] = \frac{\varphi(dv)uv}{(uv, h)t} \cdot 2^{[3|dv]} = \frac{\varphi(dv)uv}{(uv, h)t} \cdot 2^{[3|d]},$$

because $v \mid d^\infty$. Finally, replacing this degree in the expression of $\delta_\gamma^+(d)$, linearizing t , and using Lemma 10 for the equality $\delta_\gamma^+(d) = \delta_\gamma^-(d)$, we can write $\delta_\gamma(d)$ as the linear combination of the sums $S_{d,e,h}(\nu)$ given in the statement. $\square$

5. Remaining Cases

Our goal is to provide a way to write $\delta_\gamma(d)$ in closed-form when $h \neq h(1)$. We first take a quick look at the case $h = h(-1)$. Then, we study the cases $h = h(\zeta)$ for $\zeta \in \mu(K)$ different from ± 1 . This may only happen when $\Delta_K = -4$ or $\Delta_K = -3$. The next formula allows us to write $\delta_\gamma(d)$ solely in terms of $\delta_{-\gamma}(\cdot)$, which can be written in closed-form by Section 4. This is done via a formula used by Wiertelak [22] and Moree [13] that links $\text{ord}_\pi(\gamma)$ and $\text{ord}_\pi(-\gamma)$. We write $\zeta\gamma = \gamma_0^h$ for some $\gamma_0 \in K$.

Theorem 20. *For every $d \geq 2$, we have*

$$\delta_\gamma(d) = \begin{cases} \delta_{-\gamma}(2d) + \delta_{-\gamma}(d/2) - \delta_{-\gamma}(d), & \text{if } v_2(d) = 1; \\ \delta_{-\gamma}(d), & \text{otherwise.} \end{cases}$$

In particular, if $h = h(-1)$, then the values for $\delta_{-\gamma}$ can be computed using Theorems 13, 14, 15, 17, 18, or 19.

Proof. Let $p \in \mathcal{R}_\gamma(d)$ and π be a prime ideal of $\mathcal{O}_K$ lying above p . We have

$$\text{ord}_\pi(\gamma) = \begin{cases} \text{ord}_\pi(-\gamma)/2, & \text{if } 2 \nmid \text{ord}_\pi(\gamma); \\ 2 \text{ord}_\pi(-\gamma), & \text{if } v_2(\text{ord}_\pi(\gamma)) = 1; \\ \text{ord}_\pi(-\gamma), & \text{if } 4 \mid \text{ord}_\pi(\gamma). \end{cases}$$

We see that $\text{ord}_\pi(\gamma) = \text{ord}_\pi(-\gamma)$ if $4 \mid d$ and $d \mid \text{ord}_\pi(\gamma)$. Moreover, if $2 \nmid d$, then $d \mid \text{ord}_\pi(\gamma)$ if and only if $d \mid \text{ord}_\pi(-\gamma)$. Hence $\mathcal{R}_\gamma(d) = \mathcal{R}_{-\gamma}(d)$ and the result follows if $v_2(d) \neq 1$. Assume $v_2(d) = 1$. Then, $\mathcal{R}_\gamma(d)$ is the union of the sets

$$A_1 = \{p \text{ prime} : p \nmid a_2\Delta \text{ and } 2d \mid \text{ord}_\pi(\gamma)\},$$

and $A_2 = \mathcal{R}_\gamma(d) \setminus A_1$. Any $p \in A_1$ satisfies $4 \mid \text{ord}_\pi(\gamma)$, so that $\text{ord}_\pi(\gamma) = \text{ord}_\pi(-\gamma)$. Thus, we have $A_1 = \mathcal{R}_{-\gamma}(2d)$. For A_2 , we see that any p in this set satisfies

$$d \mid \text{ord}_\pi(\gamma) \quad \text{and} \quad 2d \nmid \text{ord}_\pi(\gamma).$$

It follows that $\text{ord}_\pi(\gamma) = 2\text{ord}_\pi(-\gamma)$, and that $p \in A_2$ if and only if $d \mid 2\text{ord}_\pi(-\gamma)$ and $d \nmid \text{ord}_\pi(-\gamma)$. We find that $A_2 = \mathcal{R}_{-\gamma}(d/2) \setminus \mathcal{R}_{-\gamma}(d)$. The result follows by taking the natural density of these sets, which exists by Sanna's theorem [15]. $\square$

Note that these switching formulas between γ and $-\gamma$ given in Theorem 20 remain valid for δ_γ^+ and δ_γ^- , and that the proof is the same. We also use the following:

Lemma 21. *Assume that $\Delta_K \in \{-4, -3\}$. Let $p \mid \Delta_K$ be prime and $d \geq 2$, $p \nmid d$. Then, we have*

$$[K_{p^{k+j}dv,uv} : \mathbb{Q}] = [K_{dv,uv} : \mathbb{Q}] \cdot \begin{cases} 3^{k+j-1}, & \text{if } p = 3; \\ 2^{k+j-2}, & \text{if } p = 2 \text{ and } k+j \geq 2; \\ 1, & \text{if } p = 2 \text{ and } (k,j) = (1,0), \end{cases}$$

for all $u \mid d$, $v \mid d^\infty$, $k \geq 1$, and $j \geq 0$.

Proof. It suffices to see that $K_{p^{k+j}dv,uv}$ is the compositum of $K_{dv,uv}$ and $K(\zeta_{p^{k+j}})$, which are linearly disjoint over K . Then, we use the formula

$$[K_{p^{k+j}dv,uv} : \mathbb{Q}] = [K_{dv,uv} : \mathbb{Q}] \cdot [K(\zeta_{p^{k+j}}) : K],$$

for composita, and compute the degree of the cyclotomic extension $K(\zeta_{p^{k+j}})/K$. $\square$

5.1. The case $h \neq h(1)$ and $\Delta_K = -4$

We may assume that $h = h(i) \neq h(\pm 1)$. Indeed, Theorem 20 already addresses the case $h = h(-1)$. Moreover, it allows us to switch between δ_γ and $\delta_{-\gamma}$. Thus, we may assume $h = h(i)$ in the following. Note that $2 \mid h$ because $h \neq h(\pm 1)$.

Theorem 22. *Assume $\Delta_K = -4$ and $h = h(i) \neq h(\pm 1)$. Let $d, k \geq 1$ be integers, $2 \nmid d$, and Δ_1 and $\mathfrak{f}(L)$ be as defined in Lemmas 4 and 5 for γ_0^{h/h_2} . Then,*

$$\delta_\gamma(2^k d) = \delta_{i_\gamma}(d) \cdot \begin{cases} 1 - \frac{2^k}{3 \cdot 2^{m+2} h_2}, & \text{if } k \in \{1, 2\}; \\ \frac{8}{3 \cdot 2^{k+m} h_2}, & \text{if } k \geq 3, \end{cases}$$

where $m = [\Delta_1 \mid 8d] + [\mathfrak{f}(L) \mid 16d]$ and $\delta_{i_\gamma}(d)$ is computed in [15, Theorem 1.1].

Proof. We give the main steps of the proof. First, by Lemma 10, we can write

$$(11) \quad \delta_\gamma(2^k d) = \sum_{j \geq 0} \sum_{v | d^\infty} \sum_{0 \leq l \leq 1} \sum_{u | d} \frac{2\mu(2^l u)}{[K_{2^{k+j}dv, 2^{j+l}uv} : \mathbb{Q}]}.$$

Then, by Kummer theory, we see that $K_{2^{k+j}dv, uv}$ and $K_{2^{k+j}dv, 2^{j+l}}$ are linearly disjoint over $K(\zeta_{2^{k+j}dv})$. By the formula for the degree of composita, we have

$$[K_{2^{k+j}dv, 2^{j+l}uv} : \mathbb{Q}] = [K_{2^{k+j}dv, uv} : \mathbb{Q}] \cdot [K_{2^{k+j}dv, 2^{j+l}} : K(\zeta_{2^{k+j}dv})].$$

The degree of $K_{2^{k+j}dv, uv}/\mathbb{Q}$ is computed in Lemma 21. To compute the second degree, we need to find the largest $n \geq 1$ such that $\gamma_0^h = x^{2^n}$ in $K(\zeta_{2^{k+j}dv})$. The degree is easily computed when $k+j \leq 3$. If $k+j \geq 4$, then γ_0^{h/h_2} is a square, respectively a 4-th power, in $K(\zeta_{2^{k+j}dv})$ if and only if $\Delta_1 \mid 2^{k+j}dv$, respectively $\mathfrak{f}(L) \mid 2^{k+j}dv$, by Lemmas 4 and 5. Because Δ_1 and $\mathfrak{f}(L)$ are square-free except possibly for a factor 2^n , where $n \leq 3$ for Δ_1 and $n \leq 4$ for $\mathfrak{f}(L)$, these conditions become $\Delta_1 \mid 8d$ and $\mathfrak{f}(L) \mid 16d$. Hence the definition of m . If $k+j \geq 4$, we have

$$[K_{2^{k+j}dv, 2^{j+l}} : K(\zeta_{2^{k+j}dv})] = \begin{cases} 2^{l+2-k}, & \text{if } k \in \{1, 2\}; \\ 1, & \text{if } k \geq 3, \end{cases}$$

when $j+l \leq v_2(2^m h)$. Otherwise, when $j+l > v_2(2^m h)$, we have

$$[K_{2^{k+j}dv, 2^{j+l}} : K(\zeta_{2^{k+j}dv})] = \begin{cases} [2^{l+1}, 2^{j+l-v_2(2^m h)}], & \text{if } k = 1; \\ 2^{j+l-v_2(2^m h)}, & \text{if } k \geq 2. \end{cases}$$

In the latter case, we first used [14, Lemma 4] with the fields $K(\zeta_{2^{k+j}dv}, (-i)^{1/2^{j+l}})$ and $K(\zeta_{2^{k+j}dv}, \gamma_0^{h_0/2^{j+l}})$, which are linearly disjoint over $K(\zeta_{2^{k+j}dv})$. Then, we used Theorem 2 to compute the degree of $K(\zeta_{2^{k+j}dv}, \gamma_0^{h_0/2^{j+l}})$. In particular, we find that the degree $A_{j,k,l} := [K_{2^{k+j}dv, 2^{j+l}} : K(\zeta_{2^{k+j}dv})]$ is independent of u and v . Using Lemma 21, we obtain

$$\delta_\gamma(2^k d) = \delta_\gamma(d) \cdot \sum_{j \geq 0} \left(\frac{1}{A_{j,k,0}} - \frac{1}{A_{j,k,1}} \right) \cdot \begin{cases} 1, & \text{if } k = 1 \text{ and } j = 0; \\ 2^{2-j-k}, & \text{if } k+j \geq 2. \end{cases}$$

We have now obtained $\delta_\gamma(d)$ multiplied by a geometric series, the computation of which yields the claimed formula.

Finally, note that $K(\zeta_{dv}, \gamma^{1/uv})$ and $K(\zeta_{dv}, \gamma_0^{h/uv})$ are equal for all $u \mid d$ and $v \mid d^\infty$. Indeed, since d is odd, we have $(-i)^{1/uv} = \pm i$. Therefore, $\gamma^{1/uv} = \pm i \gamma_0^{h/uv}$ and $\delta_\gamma(d) = \delta_{i\gamma}(d)$ for all odd $d \geq 1$, which is computed in [15, Theorem 1.1]. $\square$

5.2. The case $h \neq h(1)$ and $\Delta_K = -3$

We compute the Dirichlet density of $\mathcal{R}_\gamma(d)$ when $h \neq h(1)$ and $\Delta_K = -3$. We may assume that $h = h(\zeta_3) \neq h(\pm 1)$. The case $h = h(-1)$ follows from Theorem 20, which in addition, allows us to switch between δ_γ and $\delta_{-\gamma}$. In particular, when $h = h(\zeta_6)$ or $h = h(\zeta_6^{-1})$, we

can switch to $-\gamma$, for which $h = h(\zeta_3)$ or $h = h(\zeta_3^2)$ respectively. The proofs for $h = h(\zeta_3)$ and $h = h(\zeta_3^2)$ are identical, so we present them for $h = h(\omega)$, where ω is a primitive 3-rd root of unity. Note that $h \neq h(\pm 1)$ implies $3 \mid h$.

Theorem 23. *Assume $\Delta_K = -3$ and $h = h(\omega) \neq h(\pm 1)$. Let $d, k \geq 1$ be integers, $3 \nmid d$, and $\mathfrak{f}(L)$ be as defined in Lemma 6 for γ_0^{h/h_3} . Then,*

$$\delta_\gamma(3^k d) = \delta_{\omega\gamma}(d) \cdot \begin{cases} 1 - \frac{1}{4 \cdot 3^m h_3}, & \text{if } k = 1; \\ \frac{9}{4 \cdot 3^{k+m} h_3}, & \text{if } k \geq 2, \end{cases}$$

where $m = [\mathfrak{f}(L) \mid 9d]$ and $\delta_{\omega\gamma}(d)$ is computed in [15, Theorem 1.1] if $2 \nmid d$ and in Theorem 19 otherwise.

Proof. The proof is almost identical to that of Theorem 22, so we may skip a few details. First, we write $\delta_\gamma(3^k d)$ in the same way as (11) with the degree

$$[K_{3^{k+j}dv, 3^{j+lv}} : \mathbb{Q}] = [K_{3^{k+j}dv, uv} : \mathbb{Q}] \cdot [K_{3^{k+j}dv, 3^{j+l}} : K(\zeta_{3^{k+j}dv})].$$

The fields are linearly disjoint over $K(\zeta_{3^{k+j}dv})$ by Kummer theory. Hence the degree formula. The first degree is computed in Lemma 21. For the second, note that γ_0^{h/h_3} is a cube root in $K(\zeta_{3^{k+j}dv})$ if and only if $\mathfrak{f}(L) \mid 3^{k+j}dv$ by Lemma 6. However, because $\mathfrak{f}(L)$ is square-free apart from a factor of 3^n ($n \leq 2$), we have $\mathfrak{f}(L) \mid 3^{k+j}dv$ if and only if $\mathfrak{f}(L) \mid 9d$ and $j + k \geq 2$. We obtained

$$[K_{3^{k+j}dv, 3^{j+l}} : K(\zeta_{3^{k+j}dv})] = \begin{cases} 3^l, & \text{if } k = 1 \text{ and } j + l \leq v_3(3^m h); \\ 1, & \text{if } k \geq 2 \text{ and } j + l \leq v_3(3^m h); \\ 3^{j+l-v_3(3^m h)}, & \text{if } j + l > v_3(3^m h), \end{cases}$$

where we used that $K_{3^{k+j}dv, 3^{j+l}}$ is a cyclotomic field when $j + l \leq v_3(3^m h)$, and [14, Lemma 4] and Theorem 2 when $j + l > v_3(3^m h)$. The rest of the proof follows as Theorem 22. $\square$

6. Numerical data

In this last section, we present numerical demonstrations of our results. In each table, we write γ in the form $u + v\sqrt{\Delta_K}$, where $u, v \in \mathbb{Q}^\times$. For each of them, we computed h and ζ such that $\zeta\gamma = \gamma_0^h$ with $\gamma_0 \in K$. We denote by $\tilde{\gamma}$ the number $\zeta\gamma$. In Tables 1 and 2, we display the value of $\tilde{\gamma}^{1/h_2}$ from which one can compute c , K_1 , K_2 , and their discriminants Δ_1 , Δ_2 .

In Table 1, we display $\mathcal{Q} = [N_{K/\mathbb{Q}}(\tilde{\gamma}^{1/h_2})]$. In Table 2, we give the value of $\mathfrak{f}(L)$, the conductor of $\mathbb{Q}(i, \tilde{\gamma}^{1/4h_2})$. Similarly, we write explicitly the value of $\tilde{\gamma}^{1/h_6}$ in Table 3, and give the value of $\mathfrak{f}(L)$, the conductor of $\mathbb{Q}(\omega, \tilde{\gamma}^{1/3h_6})$. Note that we choose $\tilde{\gamma}^{1/h_6}$ instead

of writing $\tilde{\gamma}^{1/h_2}$ and $\tilde{\gamma}^{1/h_3}$ separately. This does not change the values Δ_1 and Δ_2 , nor of the conductor $\mathfrak{f}(L)$, thanks to Bézout's identity.

Finally, for each value of d in the tables, we computed $\delta_\gamma(d)$, which we display as a rational and in numerical form in the “num.” column, truncated to 6 decimal places. In the “exp.” column, we provide the experimental value of $\delta_\gamma(d)$ computed via SageMath [19] programs. The programs used in this work are available in the author's GitHub repository [3]. We underline the digits of the “exp.” column that coincide with those in the “num.” column. We tested primes up to 10^7 .

γ	h	ζ	$\tilde{\gamma}^{1/h_2}$	$\mathcal{Q}$	d	$\delta_\gamma(d)$	num.	exp.
$3 + \sqrt{8}$	2	1	$\frac{2+\sqrt{8}}{2}$	0	6	17/64	0.265625	<u>0.265670</u>
$3 + \sqrt{8}$	2	1	$\frac{2+\sqrt{8}}{2}$	0	20	25/288	0.086805	<u>0.086782</u>
$\frac{-27-5\sqrt{29}}{2}$	2	-1	$\frac{5+\sqrt{29}}{2}$	0	8	1/6	0.166666	<u>0.166473</u>
$\frac{-27-5\sqrt{29}}{2}$	2	-1	$\frac{5+\sqrt{29}}{2}$	0	10	5/36	0.138888	<u>0.139166</u>
$\frac{17+7\sqrt{-15}}{32}$	4	1	$\frac{1-\sqrt{-15}}{4}$	1	10	5/288	0.017361	<u>0.017287</u>
$\frac{17+7\sqrt{-15}}{32}$	4	1	$\frac{1-\sqrt{-15}}{4}$	1	30	5/384	0.013020	<u>0.013017</u>

TABLE 1. The case $\Delta_K \notin \{-4, -3\}$.

γ	h	ζ	$\tilde{\gamma}^{1/h_2}$	$\mathfrak{f}(L)$	d	$\delta_\gamma(d)$	num.	exp.
$\frac{-3+2\sqrt{-4}}{5}$	1	1	$\frac{-3+2\sqrt{-4}}{5}$	20	8	1/3	0.333333	<u>0.333427</u>
$\frac{-3+2\sqrt{-4}}{5}$	1	1	$\frac{-3+2\sqrt{-4}}{5}$	20	10	5/72	0.069444	<u>0.069279</u>
$\frac{48+7\sqrt{-4}}{50}$	2	i	$\frac{3+2\sqrt{-4}}{5}$	40	10	235/1152	0.203993	<u>0.203844</u>
$\frac{48+7\sqrt{-4}}{50}$	2	i	$\frac{3+2\sqrt{-4}}{5}$	40	24	1/16	0.062500	<u>0.062553</u>
$\frac{-240-119\sqrt{-4}}{338}$	2	i	$\frac{-24+5\sqrt{-4}}{26}$	208	26	661/8064	0.075768	<u>0.075771</u>
$\frac{-240-119\sqrt{-4}}{338}$	2	i	$\frac{-24+5\sqrt{-4}}{26}$	208	28	35/288	0.121527	<u>0.121457</u>

TABLE 2. The case $\Delta_K = -4$.

γ	h	ζ	$\tilde{\gamma}^{1/h_6}$	$\mathfrak{f}(L)$	d	$\delta_\gamma(d)$	num.	exp.
$\frac{-13+3\sqrt{-3}}{14}$	1	1	$\frac{-13+3\sqrt{-3}}{14}$	7	3	3/4	0.750000	<u>0.750061</u>
$\frac{-13+3\sqrt{-3}}{14}$	1	1	$\frac{-13+3\sqrt{-3}}{14}$	7	14	35/288	0.121527	<u>0.121231</u>
$\frac{683+37\sqrt{-3}}{686}$	3	ω^2	$\frac{1+4\sqrt{-3}}{7}$	63	9	1/12	0.083333	<u>0.083407</u>
$\frac{683+37\sqrt{-3}}{686}$	3	ω^2	$\frac{1+4\sqrt{-3}}{7}$	63	42	1225/10368	0.118152	<u>0.117806</u>
$\frac{1031-520\sqrt{-3}}{1369}$	2	-1	$\frac{13+20\sqrt{-3}}{37}$	333	6	5/8	0.625000	<u>0.624809</u>
$\frac{1031-520\sqrt{-3}}{1369}$	2	-1	$\frac{13+20\sqrt{-3}}{37}$	333	111	407/16416	0.024792	<u>0.024823</u>

TABLE 3. The case $\Delta_K = -3$.

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References

- [1] C. Ballot. Lucas sequences with cyclotomic root field. *Dissertationes Math.*, 490:92, 2013. <https://doi.org/10.4064/dm490-0-1>.
- [2] A. Cavallo. Denesting cubic radicals, 2024. <https://arxiv.org/abs/2403.04776>.
- [3] J. Cera Da Conceição. GitHub Repository,. Available at <https://github.com/jceradaconceicao/sage-programs>.
- [4] P. Cubre and J. Rouse. Divisibility properties of the Fibonacci entry point. *Proc. Amer. Math. Soc.*, 142(11):3771–3785, 2014. <https://doi.org/10.1090/S0002-9939-2014-12269-6>.
- [5] G. Gras. *Class Field Theory: From Theory to Practice*. Springer-Verlag, Berlin, 2003. Translated from the French by Henri Cohen.
- [6] H. Hasse. Arithmetische Theorie der kubischen Zahlkörper auf klassenkörpertheoretischer Grundlage. *Math. Z.*, 31(1):565–582, 1930. <https://doi.org/10.1007/BF01246435>.
- [7] H. Hasse. Über die Dichte der Primzahlen p , für die eine vorgegebene ganzrationale Zahl $a \neq 0$ von durch eine vorgegebene Primzahl $l \neq 2$ teilbarer bzw. unteilbarer Ordnung mod. p ist. *Math. Ann.*, 162:74–76, 1965/66. <https://doi.org/10.1007/BF01361933>.
- [8] H. Hasse. Über die Dichte der Primzahlen p , für die eine vorgegebene ganzrationale Zahl $a \neq 0$ von gerader bzw. ungerader Ordnung mod. p ist. *Math. Ann.*, 166:19–23, 1966. <https://doi.org/10.1007/BF01361432>.
- [9] J. G. Huard, B. K. Spearman, and K. S. Williams. A short proof of the formula for the conductor of an abelian cubic field. *Skr. K. Nor. Vidensk. Selsk.*, 2:1–8, 1994.
- [10] G. Karpilovsky. *Topics in Field Theory*. North-Holland Publishing Co., Amsterdam, 1989. North-Holland Mathematics Studies, 155.

- [11] J. C. Lagarias. The set of primes dividing the Lucas numbers has density $2/3$. *Pacific J. Math.*, 118(2):449–461, 1985. <http://projecteuclid.org/euclid.pjm/1102706452>.
- [12] Y. Luo, J. Hong, and Y. Liu. The Dirichlet density of set of primes related to the Lucas sequences. *Int. J. Number Theory*, 22(3):461–487, 2026. <https://doi.org/10.1142/S1793042126500284>.
- [13] P. Moree. On primes p for which d divides $\text{ord}_p(g)$. *Funct. Approx. Comment. Math.*, 33:85–95, 2005. <https://doi.org/10.7169/facm/1538186603>.
- [14] A. Perucca. The order of the reductions of an algebraic integer. *J. Number Theory*, 148:121–136, 2015. <https://doi.org/10.1016/j.jnt.2014.09.022>.
- [15] C. Sanna. On the divisibility of the rank of appearance of a Lucas sequence. *Int. J. Number Theory*, 18(10):2145–2156, 2022. <https://doi.org/10.1142/S1793042122501093>.
- [16] C. Sanna. On the index of appearance of a Lucas sequence. *Ramanujan J.*, 63(4):1199–1223, 2024. <https://doi.org/10.1007/s11139-023-00811-4>.
- [17] A. Schinzel. A refinement of a theorem of Gerst on power residues. *Acta Arith.*, 17:161–168, 1970. <https://doi.org/10.4064/aa-17-2-161-168>.
- [18] B. K. Spearman and K. S. Williams. The conductor of a cyclic quartic field using Gauss sums. *Czechoslovak Math. J.*, 47(122)(3):453–462, 1997. <https://doi.org/10.1023/A:1022407300351>.
- [19] The Sage Developers. *SageMath, the Sage Mathematics Software System (Version 10.7)*, 2025. <https://www.sagemath.org>.
- [20] K. Wiertelak. On the density of some sets of primes. I. *Acta Arith.*, 34(3):183–196, 1977/78. <https://doi.org/10.4064/aa-34-3-183-196>.
- [21] K. Wiertelak. On the density of some sets of primes. II. *Acta Arith.*, 34(3):197–210, 1977/78. <https://doi.org/10.4064/aa-34-3-197-210>.
- [22] K. Wiertelak. On the density of some sets of primes. IV. *Acta Arith.*, 43(2):177–190, 1984. <https://doi.org/10.4064/aa-43-2-177-190>.

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